Student dropout warnings.

Who gets missed by
the same warning rule?

University of Edinburgh

An enrolment model and its missed warnings

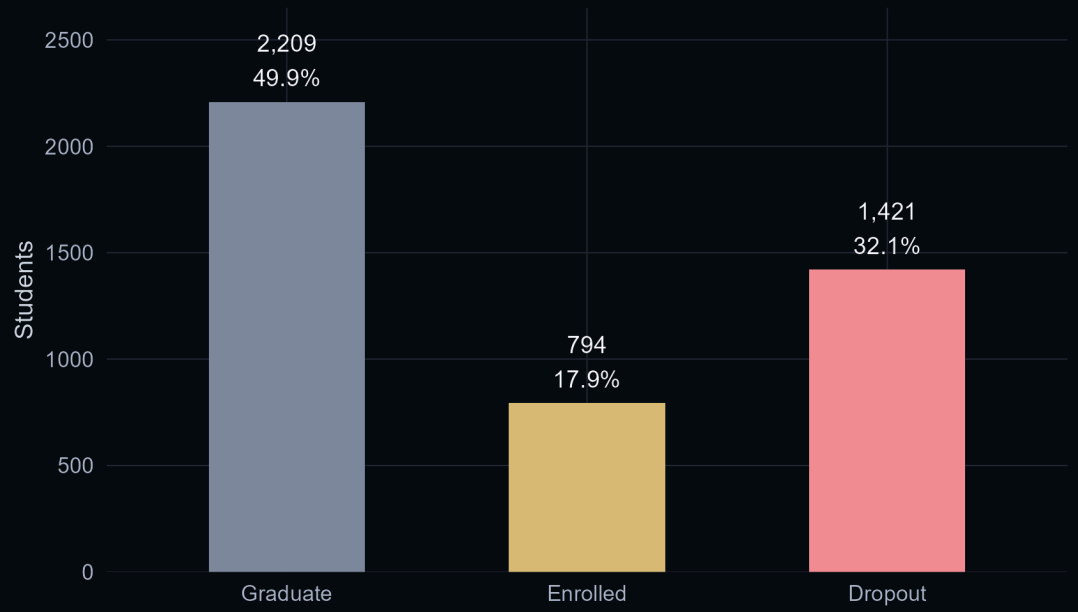
UCI 697 / ONE PORTUGUESE INSTITUTION
Analysis 05 NOV 2026 / Revision 11 NOV 2026

What counts as recorded dropout?

4,424

student records.
1,421 recorded dropouts.

A course or institution change also counts as **Dropout**.



Target: recorded dropout by the normal course endpoint.

Graduate and Enrolled mean no recorded dropout then. Enrolled does not mean graduated.

Realinho et al. (2022), pp. 1-4. Enrolments 2008/09-2018/19; endpoint usually 3 years, 4 for Nursing. No causal or Edinburgh-wide claim.

When would we need to make the prediction?

The prediction starts at enrolment

AVAILABLE AT ENROLMENT

Age, grade and course

Inputs to the original model

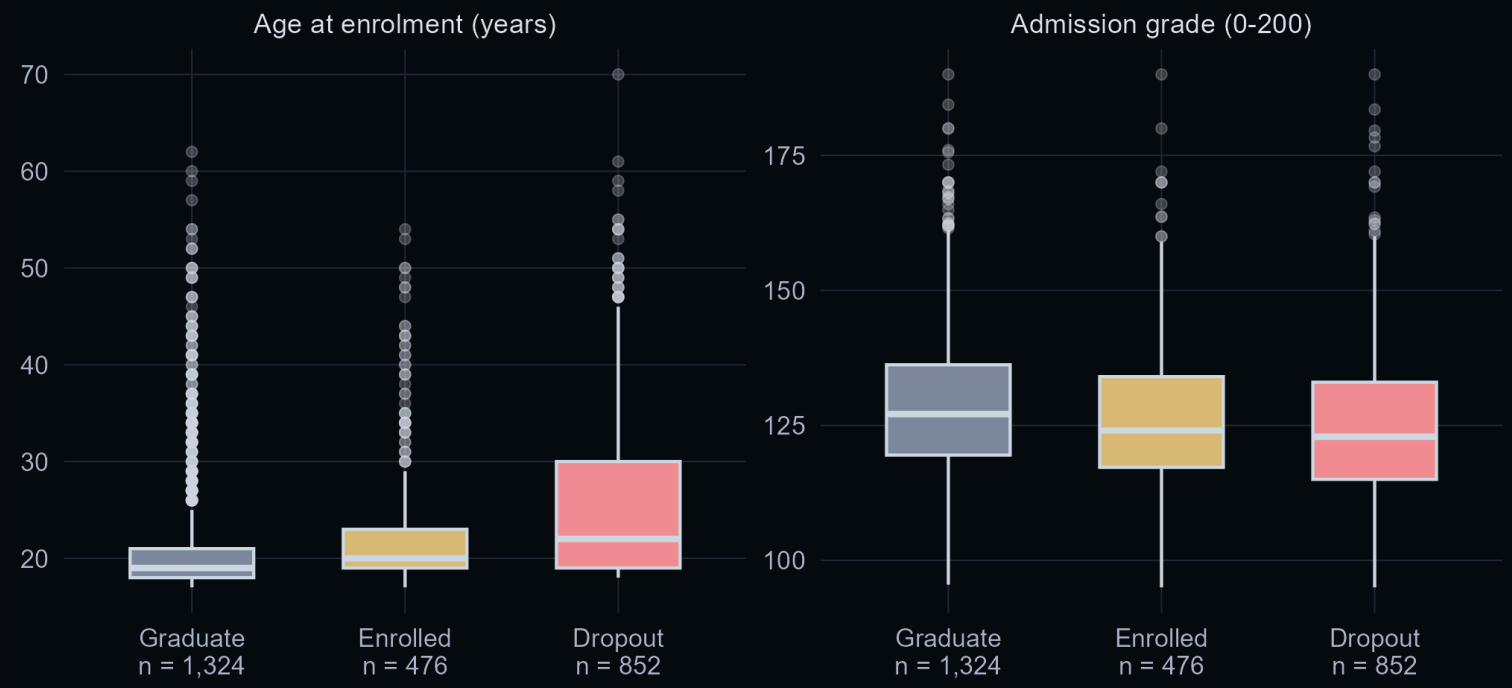
OBSERVED AT THE COURSE ENDPOINT

Recorded outcome

Dropout, Enrolled or Graduate

How much do the available inputs tell us?

What do we know at enrolment?



Higher median age in the recorded-dropout group.

Admission grades overlap substantially. No individual outcome is determined.

Training only, n = 2,652; associations, not causes.

Quiz 7 recommended Data to Viz; this prompted our figure review. Boxplot advice.

How do we combine these overlapping signals into one risk score?

From enrolment to a warning

Age, admission grade and course → Estimated dropout risk → Warn if risk reaches the threshold

01 / FIT

2,652

Training

Fit the model.

02 / CHOOSE

883

Validation

Choose one warning threshold.

03 / EVALUATE

889

Test

Evaluate the frozen rule.

Course is categorical; age and grade are additive on the log-odds scale, without interactions.

Later and timing-uncertain inputs are excluded. Recorded gender is reserved for the audit.

> The regression lectures help us review these choices. Logistic regression required further learning.

Seed 20261105; approximately 60/20/20, stratified by original outcome and recorded gender. Model fitting and validation use no test outcomes.

What score should trigger a warning?

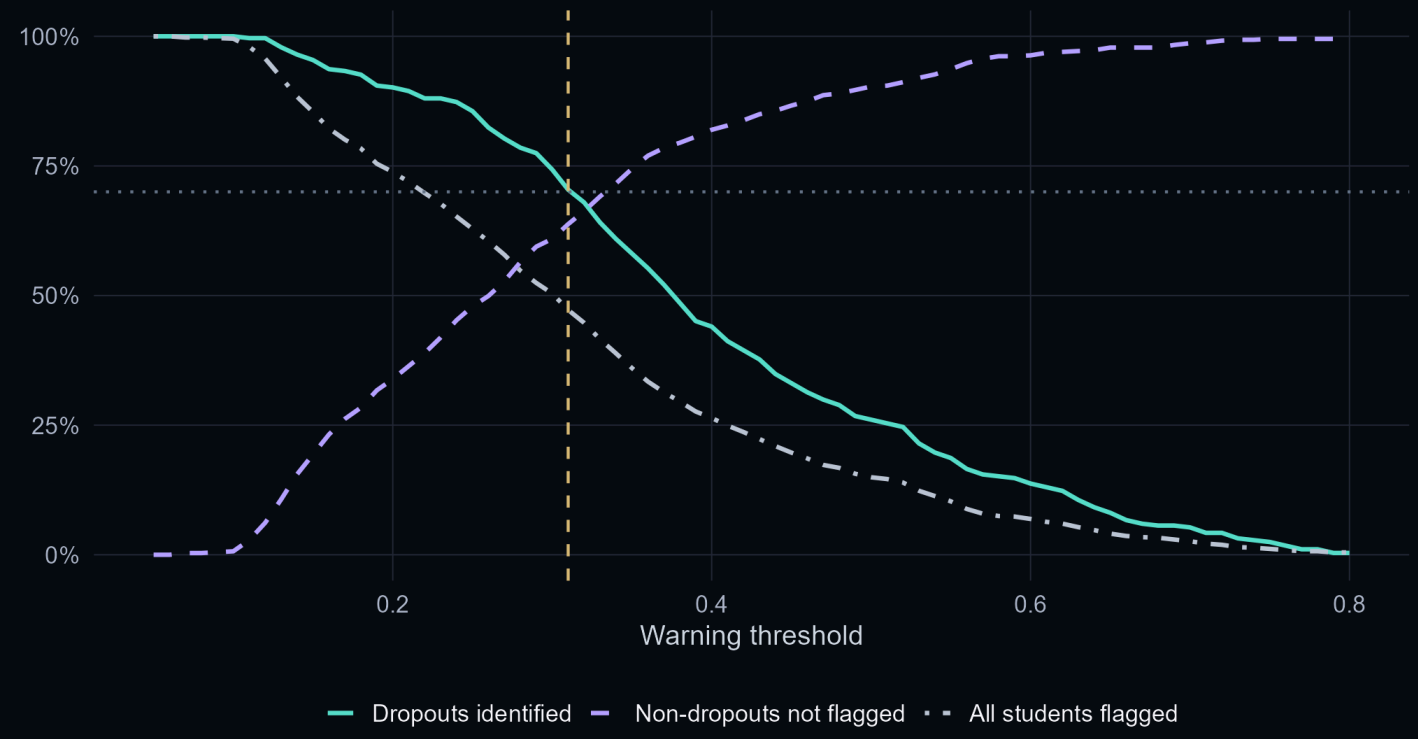
What score triggers a warning?

0.31

One common threshold.

Highest specificity while identifying at least **70%** of validation dropouts.

Validation sensitivity:
200 / 284 = 70.4%



Our illustrative target, not a university policy. Grid 0.05-0.80 in steps of 0.01; one threshold for both groups. We then froze the rule and evaluated its sensitivity, precision and errors on test records.

What changes if we warn at 0.31 instead of 0.50?

A lower threshold changes who gets warned

MORE DROPOUT RECORDS FOUND

+107

187 at 0.31 versus 80 at 0.50

MORE FALSE WARNINGS

+177

222 at 0.31 versus 45 at 0.50

What did the chosen 0.31 rule achieve overall?

How many dropouts did we miss?

65.6%
Sensitivity / recall
187 / 285 dropout records found

45.7%
Precision / PPV
187 / 409 warnings matched

64.0%
Accuracy
569 / 889 decisions matched

RULE	SENSITIVITY	PRECISION (PPV)	ACCURACY
Always no warning	0.0%	Not defined	67.9%
Threshold 0.50	28.1%	64.0%	71.9%
Validation rule: 0.31	65.6%	45.7%	64.0%

The rule missed **98 recorded dropouts**. Who were they?

Test n = 889; 222 false warnings; 409 / 889 (46.0%) flagged. Sensitivity 95% conditional bootstrap interval: 60.3% to 71.0%. A false warning is relative to the recorded label.

The missed-dropout rate differs by recorded gender

RECORDED FEMALE

44.4%

64 of 144 recorded dropouts missed

RECORDED MALE

24.1%

34 of 141 recorded dropouts missed

Female minus male: **20.3 percentage points** (95% interval: 9.5 to 30.7).

Who was more often warned without a recorded dropout?

Among students with no recorded dropout in the test set

32.0%

RECORDED FEMALE

138 of 431 students flagged

48.6%

RECORDED MALE

84 of 173 students flagged

Do we see a similar trade-off when we group students by whether they left home to study?

Same 0.31 threshold and test set as the missed-dropout comparison.

Do the error patterns extend beyond gender?

Displaced = 1 means recorded as having left home to study. The dataset codes this as a yes/no field.

THRESHOLD	DISPLACED	MISSED DROPOUT	FALSE WARNING
0.31	No (0)	39/146 (26.7%)	105/236 (44.5%)
0.31	Yes (1)	59/139 (42.4%)	117/368 (31.8%)
0.50	No (0)	90/146 (61.6%)	35/236 (14.8%)
0.50	Yes (1)	115/139 (82.7%)	10/368 (2.7%)

At 0.31, students marked displaced had a **15.7 percentage-point higher miss rate** and a lower false-warning rate.

Across both groupings, these errors give us a reason to review warnings before offering support.

Review the warning before offering support

A missed warning can delay an offer of support. A false warning can prompt an unnecessary follow-up.

Check in with the student.

Confirm the record, ask what support they need and offer help.

Keep support open.

Students who received no warning can still ask for help.

Check capacity and validate on later cohorts.

In the test set, 46.0% of students were flagged.

This analysis did not test whether outreach prevents dropout.

Historical records from one institution. The rule still needs evaluation on later cohorts.

Sources and assistance

- Realinho et al. **UCI dataset 697**, DOI 10.24432/C5MC89. CC BY 4.0. Retrieved 5 November 2026.
- Realinho et al. (2022). **Predicting Student Dropout and Academic Success**, *Data* 7, 146. 10.3390/data7110146.
- IDS course materials: **Algorithmic Bias**, pp. 6, 16-19; **Quantifying uncertainty**, pp. 21, 39, 46-53.
- IDS regression lectures: **Introducing linear models**, pp. 22-28, 48-54; **Model checking and multiple linear regression**, pp. 24-41, 57-63.
- R documentation: binomial logistic regression with glm. Fairness definitions: Fairness and Machine Learning.
- Figure review: Data to Viz, prompted by Quiz 7; page-specific advice is recorded in the notes.
- Review tools: Sciwrite (Barba); Anti-defensive Writing; Kassis et al. (2026), Scientific Agent Skills.

We applied our course methods to data handling, plots, bootstrap and the fairness audit.
For unfamiliar methods, we used ChatGPT and documentation, then rewrote and checked the main R analysis.
ChatGPT also supported initial drafts, source checks and the revision of slide titles and transitions. We take responsibility for the final work.

The six photographs are AI-generated illustrations of campus, enrolment and student support scenarios.